

SF2745 Advanced Complex Analysis: Oral Exam Guide

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Chapter 1 - Preliminaries

Definition: (**Stereographic projection**)

Identifying the complex plane $\mathbb{C}$ with the plane $\{(x, y, 0) \in \mathbb{R}^3\}$, let the **stereographic projection** z^* of $z \in \mathbb{C} \cup \{\infty\}$ be the point on the unit sphere in $\mathbb{R}^3$ which lies on the line segment from the north pole to z .

↳ Motivation: Useful to compactify $\mathbb{C} \cup \{\infty\}$ and to measure distances to ∞ .

Theorem:

Under stereographic projection, circles and lines in $\mathbb{C}$ correspond to circles in S^2 .

↳ Motivation: Working on $\mathbb{C} \cup \{\infty\}$ is similar to working on S^2 . For example, LFTs on $\mathbb{C} \cup \{\infty\}$ are precisely the analytic automorphisms of S^2 .

Chapter 2 - Analytic Functions

Definition: (**Power series**)

A **convergent power series centered at z_0** is a series of the form

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

that converges on $|z - z_0| < r$.

↳ Motivation: Power series always converge on disks.

Theorem: (**Weierstrass M-test**)

If $|a_n(z - z_0)^n| \leq M_n$ for $|z - z_0| \leq r$ and $\sum M_n < \infty$, then $\sum a_n(z - z_0)^n$ converges uniformly and absolutely on $|z - z_0| \leq r$.

↳ Proof:

Look at $|S_N - S_M|$.

Let $S_N(z) := \sum_{n=0}^N a_n(z - z_0)^n$. For $N < M$ and $|z - z_0| \leq r$,

$$|S_N(z) - S_M(z)| = \left| \sum_{n=N+1}^M a_n(z - z_0)^n \right| \leq \sum_{n=N+1}^M M_n \rightarrow 0 \quad \text{as } N, M \rightarrow \infty$$

since this is the tail of the convergent series $\sum M_n$. Hence $\{S_N\}$ is a Cauchy sequence (independent of z) and thus converges uniformly. Replace S_N with $\sum_{n=0}^N |a_n(z - z_0)^n|$ to show absolute convergence.

Theorem: (**Root test**)

For a formal power series $\sum a_n(z - z_0)^n$, let

$$R := 1/\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \in [0, \infty].$$

Then the series

- (a) converges absolutely in $|z - z_0| < R$;
- (b) converges uniformly in $|z - z_0| \leq r < R$;
- (c) diverges on $|z - z_0| > R$.

↳ Motivation: Shows that power series converge on disks, and finds the radius of convergence.

Definition: (**Radius of convergence**)

The root test gives a value R called the **radius of convergence**.

Definition: (**Analytic**)

A function f is **analytic** at z_0 if $f(z) = \sum a_n(z - z_0)^n$ in some disk at z_0 .

↳ Motivation: These are the main objects of study in complex analysis.

Definition: (**Region**)

A **region** is an open connected set in $\mathbb{C}$.

↳ Motivation: Analytic functions typically have regions as their domain, which are open since analyticity is local, and connected to avoid splitting up the function.

Theorem: (**Uniqueness theorem**)

If f is analytic on Ω then either $f \equiv 0$ or the zeros of f are isolated.

↳ Proof:

Show that $E = \{\text{non-isolated zeros}\}$ is closed via the complement, and open through a previous result.

Let $E := \{\text{non-isolated zeros of } f\}$. Consider $a \in \Omega \setminus E$. If a is a zero, it is isolated, and therefore contained in a neighborhood free from other zeros and thus lying in $\Omega \setminus E$. If a is not a zero, continuity gives that f is nonzero in a neighborhood thus lying in $\Omega \setminus E$. Hence E is closed. We previously showed that f is constant zero close to non-isolated zeros. Hence E is open. By connectedness, $E = \Omega$ or $E = \emptyset$.

Theorem: (**Power series representation of analytic functions**)

If f is equal to a power series on some disk $|z - z_0| < r$, then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

on that disk.

↳ Proof:

We may differentiate termwise.

If $f(z) = \sum a_n(z - z_0)^n$ on a disk D , by differentiating termwise,

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

Evaluation at z_0 gives $f'(z_0) = a_1$. By induction, $f^{(n)}(z_0) = n! a_n$.

Chapter 3 - The Maximum Principle

Theorem: (**Maximum principle**)

If f is analytic on Ω and $|f|$ attains its supremum on Ω , then f is constant.

↳ Motivation: E.g. shows that $|f|$ cannot have local maxima, and that $|f|$ is controlled by its boundary values.

Definition: (**Mean-value property**)

A function f has the **mean-value property** at z_0 if for sufficiently small r ,

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

↳ Motivation: Analytic functions have the mean-value property.

Theorem:

A non-constant analytic map on a region is an open map.

↳ Motivation: Restricts analytic maps; for example, the image cannot be $\mathbb{R}$.

Definition: (**Locally conformal**)

A function f is **locally conformal** at z_0 if it preserves angles (including direction) between curves.

↳ Motivation: Implied by conformality.

Theorem: (**Liouville's theorem**)

Every bounded entire function is constant.

↳ Proof:

Use $g(z) = \frac{f(z)-f(0)}{z}$.

For a bounded entire function f , let $g(z) := \frac{f(z)-f(0)}{z}$ which is entire and approaches 0 as $z \rightarrow \infty$. By the maximum principle, $g \equiv 0$.

Theorem: (**Schwarz's lemma**)

Suppose $f : \mathbb{D} \rightarrow \overline{\mathbb{D}}$ and $f(0) = 0$. Then,

$$|f(z)| \leq |z| \quad \text{and} \quad |f'(0)| \leq 1.$$

Moreover, if equality holds in one of the above (for $z \neq 0$), then $f(z) = cz$ where $|c| = 1$.

↳ Proof:

Use $g(z) = \frac{f(z)}{z}$.

Let $g(z) := f(z)/z$ which is analytic on $\mathbb{D}$. For $0 < r < 1$, $\sup_{|z|=r} |g(z)| \leq 1/r$. By the maximum principle, $|g(z)| \leq 1/r$ for $|z| < r$. Letting $r \rightarrow 1^-$ we get $|g(z)| \leq 1$ for $z \in \mathbb{D}$. Hence $|f(z)| \leq |z|$. Also, $g(0) = f'(0)$, showing $|f'(0)| \leq 1$.

If $z_0 \neq 0$ and $|f(z_0)| = |z_0|$, then $|g(z_0)| = 1$, so by the maximum principle, $g \equiv c$ with $|c| = 1$. Hence $f(z) = cz$. If $|f'(0)| = 1$, then $|g(0)| = 1$, so the same conclusion holds.

Definition: (**Non-tangential convergence**)

Let $f : \mathbb{D} \rightarrow \mathbb{C}$ and $\zeta \in \partial\mathbb{D}$. By **non-tangential convergence** at ζ , we mean that for every Stolz angle $\Gamma_\alpha(\zeta) := \{z \in \mathbb{D} : |z - \zeta| \leq \alpha(1 - |z|)\}$ where $\alpha > 1$,

$$\lim_{z \in \Gamma_\alpha(\zeta) \rightarrow \zeta} f(z)$$

exists.

↳ Motivation: Abel's limit theorem says that, given some convergence criteria, non-tangential convergence exists and gives what one expects of a power series.

Chapter 4 - Integration and Approximation

Definition: (**Curve integral**)

For a piecewise continuously differentiable curve $\gamma : [a, b] \rightarrow \mathbb{C}$ and a function f continuous on γ , we define

$$\int_\gamma f(z) \, dz := \int_a^b f(\gamma(t)) \gamma'(t) \, dt.$$

↳ Motivation: This is like a change of variables. Also, it is independent of parametrization.

Definition: (**Curve length**)

For a piecewise continuously differentiable curve $\gamma : [a, b] \rightarrow \mathbb{C}$, its **length** is defined as

$$\int_\gamma |dz| = \int_a^b |\gamma'(t)| \, dt.$$

↳ Motivation: We can view $\gamma'(t)$ as a velocity vector. Also, $\left| \int_\gamma f \right| \leq |\gamma| \sup_\gamma |f|$.

Definition: (**Holomorphic**)

A function f is **holomorphic** on an open set U if f' exists and is continuous on U . It is holomorphic on an arbitrary set if it is holomorphic on some larger open set.

↳ Motivation: Continuity of f' is not necessary. Will be equivalent to analyticity.

Theorem:

$$\frac{1}{2\pi i} \int_{C_r(z_0)} \frac{1}{z - a} \, dz = \begin{cases} 1, & |z_0 - a| < r, \\ 0, & |z_0 - a| > r. \end{cases}$$

↳ Proof:

Do the integral and use a geometric series.

Suppose $|z_0 - a| < r$. Then

$$\begin{aligned} \frac{1}{2\pi i} \int_C \frac{1}{z - a} \, dz &= \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{z_0 + re^{it} - a} re^{it} \, dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{1 - \frac{a - z_0}{re^{it}}} \, dt = \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} \left(\frac{a - z_0}{re^{it}} \right)^n \, dt \\ &= \sum_{n=0}^{\infty} \left(\frac{a - z_0}{r} \right)^n \frac{1}{2\pi} \int_0^{2\pi} e^{-int} \, dt = 1. \end{aligned}$$

Here the series $\sum_{n=0}^{\infty} \left(\frac{a-z_0}{re^{it}}\right)^n$ was uniformly convergent in t since $\left|\frac{a-z_0}{re^{it}}\right| = \left|\frac{a-z_0}{r}\right| < 1$. If instead $|z_0 - a| > r$, then

$$\begin{aligned} \frac{1}{2\pi i} \int_C \frac{1}{z-a} dz &= \frac{1}{2\pi} \int_0^{2\pi} \frac{-re^{it}}{a-z_0} \frac{1}{1 - \frac{re^{it}}{a-z_0}} dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{-re^{it}}{a-z_0} \sum_{n=0}^{\infty} \left(\frac{re^{it}}{a-z_0}\right)^n dt \\ &= \frac{-r}{a-z_0} \sum_{n=0}^{\infty} \left(\frac{r}{a-z_0}\right)^n \int_0^{2\pi} e^{i(n+1)t} dt = 0. \end{aligned}$$

Theorem: (**Cauchy's integral formula**)

If f is holomorphic on $|z - z_0| \leq r$ then for $|z - z_0| < r$,

$$f(z) = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

↳ Proof:

Differentiate $f(z + t(\zeta - z))$ to find an expression for $f(\zeta) - f(z)$. Then compute $\int_C \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = 0$.

Notice $f(\zeta) - f(z) = \int_0^1 \frac{d}{dt} f(z + t(\zeta - z)) dt = \int_0^1 f'(z + t(\zeta - z))(\zeta - z) dt$. Therefore,

$$\begin{aligned} \int_C \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta &= \int_C \int_0^1 f'(z + t(\zeta - z)) dt d\zeta = \int_0^1 \int_C f'(z + t(\zeta - z)) d\zeta dt \\ &= \lim_{\varepsilon \rightarrow 0^+} \int_{\varepsilon}^1 \int_C \frac{d}{d\zeta} f(z + t(\zeta - z)) d\zeta \frac{1}{t} dt = 0. \end{aligned}$$

Hence

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \int_C \frac{1}{\zeta - z} d\zeta = 2\pi i f(z).$$

Theorem:

A function is holomorphic on Ω if and only if it is analytic of Ω . Moreover, its series expansion at $z_0 \in \Omega$ converges in the largest disk at z_0 contained in Ω .

Theorem: (**Cauchy's integral formula for derivatives; Cauchy's estimate**)

If f is analytic on $|z - z_0| \leq r$, then

$$\frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

and

$$\left| \frac{f^{(n)}(z_0)}{n!} \right| \leq \frac{1}{r^n} \sup_{C_r(z_0)} |f|.$$

↳ Proof:

Use integral formula; expand $\frac{1}{\zeta - z}$; read off derivative.

For z close to z_0 ,

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{1}{\zeta - z_0 - (z - z_0)} f(\zeta) d\zeta = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{1}{\zeta - z_0} \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} f(\zeta) d\zeta \\ &= \frac{1}{2\pi i} \int_{C_r(z_0)} \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}} f(\zeta) d\zeta = \sum_{n=0}^{\infty} \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta (z - z_0)^n. \end{aligned}$$

From this expansion we read off the n th derivative as

$$\frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Estimating this integral gives

$$\frac{f^{(n)}(z_0)}{n!} \leq \frac{1}{2\pi} 2\pi r \sup_{\zeta \in C_r(z_0)} \left| \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \right| = \frac{1}{r^n} \sup_{C_r(z_0)} |f|.$$

Theorem: (**Morera's theorem**)

If f is continuous in an open disk B , and if

$$\int_{\partial R} f(\zeta) d\zeta = 0$$

for all closed rectangles $R \subset B$ with sides parallel to the axes, then f is analytic on B .

↳ Motivation: Gives a good way to detect analyticity since integrals often are easy to apply.

Theorem: (**Runge's theorem**)

If f is analytic on a compact set K , then for all $\varepsilon > 0$ there exists a rational function r so that

$$\sup_{z \in K} |f(z) - r(z)| < \varepsilon.$$

↳ Motivation: Useful to approximate f . We can also do it by placing the poles of r_n outside a region Ω .

Theorem: (**Weierstrass' theorem**)

Let $\{f_n\}$ be a sequence of analytic functions on Ω such that $f_n \rightarrow f$ uniformly on compact subsets. Then, f is analytic on Ω , and $f'_n \rightarrow f'$ uniformly on compact subsets.

↳ Proof:

Take limits inside integral formula.

Let B be an open disk with $\overline{B} \subset \Omega$. Then

$$f_n(z) = \frac{1}{2\pi i} \int_{\partial B} \frac{f_n(\zeta)}{\zeta - z} d\zeta$$

for all n and $z \in B$. Since $f_n \rightarrow f$ uniformly on ∂B , we get that f is continuous on ∂B and

$$f(z) = \frac{1}{2\pi i} \int_{\partial B} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Hence

$$f'(z) = \frac{1}{2\pi i} \int_{\partial B} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta$$

so f is analytic in B , and $f'_n \rightarrow f'$ uniformly on compact subsets of B . Finally any compact set can be covered by finitely many such B s.

Chapter 5 - Cauchy's Theorem

Theorem: (**Cauchy's integral formula**)

Suppose γ is a cycle contained in Ω , and suppose

$$\int_{\gamma} \frac{1}{\zeta - a} d\zeta = 0$$

for all $a \notin \Omega$. Then if f is analytic on Ω and $z \in \mathbb{C} \setminus \gamma$,

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \int_{\gamma} \frac{1}{\zeta - z} d\zeta.$$

In other words: if $\gamma \sim 0$ in Ω , then $\frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \cdot n(\gamma, z)$.

↳ Motivation: Like the simple version of Cauchy's integral formula, but generalized to cycles. The first condition ensures that Ω has no holes.

Definition: (**Index/Winding number**)

Let γ be a cycle. Its **index/winding number** about $a \notin \gamma$ is

$$n(\gamma, a) := \frac{1}{2\pi i} \int_{\gamma} \frac{1}{\zeta - a} d\zeta.$$

↳ Motivation: This counts how many times γ winds around a in the positive direction.

Definition: (**Homologous**)

Two closed curves γ_1, γ_2 are **homologous** in γ if $n(\gamma_1 - \gamma_2, a) = 0$ for all $a \notin \Omega$.

↳ Motivation: This means that γ_1 and γ_2 wind around holes in Ω an equal amount of times.

Theorem: (**Primitive function and logarithm on a simply connected region**)

Let f be analytic on a simply connected region Ω . Then,

- (a) for all closed curves $\gamma \subset \Omega$, $\int_{\gamma} f = 0$;
- (b) there exists an analytic F such that $F' = f$ on Ω ;
- (c) if $f \neq 0$, there exists an analytic g such that $f = e^g$ on Ω .

Theorem: (**Laurent series**)

Suppose f is analytic on the annulus $r < |z - z_0| < R$. Then

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

uniquely on the annulus. The series converges uniformly and absolutely on compact subsets, and

$$a_n = \frac{1}{2\pi i} \int_{C_s(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

for $r < s < R$.

Definition: (**Removable singularity/Zero/Pole/Essential singularity**)

Suppose f is analytic near z_0 . Then we may expand f in a Laurent series $f(z) = \sum a_n(z - z_0)^n$.

- (a) If $a_n = 0$ for all $n < 0$, then f has a **removable singularity** at z_0 .
- (b) If $n_0 > 0$ is the smallest index for which a_n is nonzero, then f has a **zero of order n_0** at z_0 .
- (c) If $n_0 < 0$ is the largest (in modulus) index for which a_n is nonzero, then f has a **pole of order n_0** at z_0 .
- (d) If a_n is nonzero for infinitely many $n < 0$, then f has an **essential singularity** at z_0 .

Definition: (**Meromorphic**)

If f is analytic in a region Ω except at isolated poles, then f is **meromorphic** in Ω .

Theorem: (**Casorati-Weierstrass**)

If f is analytic in $U = \{0 < |z - z_0| < r\}$ and z_0 is an essential singularity, then $f(U)$ is dense in $\mathbb{C}$.

↳ Proof:

BWOC, extend $\frac{1}{f(z)-A}$.

Suppose $f(U)$ is not dense. Then there exists $A \in \mathbb{C}$ and $\varepsilon > 0$ such that $|f(z) - A| > \varepsilon$ for all $z \in U$. Then $\frac{1}{f(z)-A}$ is analytic and bounded by $1/\varepsilon$ on U , so it extends analytically to $U \cup \{z_0\}$. Therefore its zeros are isolated, so $f - A$ is meromorphic on $U \cup \{z_0\}$, and so is f which contradicts the essential singularity.

Theorem: (**Argument principle**)

Suppose f is meromorphic in Ω with zeros $\{z_j\}$ and poles $\{p_k\}$. Suppose γ is a cycle in Ω which does not cross any zero or pole, and $\gamma \sim 0$. Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_j n(\gamma, z_j) - \sum_k n(\gamma, p_k).$$

↳ Proof:

Factor out $(z - b)^k$ to make f'/f analytic.

The sums above are finite, since $\{z_j\}$ and $\{p_k\}$ cannot cluster in Ω , and $n(\gamma, z) = 0$ if z is close to $\partial\Omega$. Write $\Omega_1 := \Omega \setminus \{z_j : n(\gamma, z_j) = 0\} \cup \{p_k : n(\gamma, p_k) = 0\}$. If b is a zero or pole of f , then there exists $k \in \mathbb{Z}$ such that

$$f(z) = (z - b)^k g(z)$$

where g is analytic close to b and $g(b) \neq 0$. Then

$$\frac{f'(z)}{f(z)} = \frac{k(z - b)^{k-1}g(z) + (z - b)^k g'(z)}{(z - b)^k g(z)} = \frac{k}{z - b} + \frac{g'(z)}{g(z)}$$

where g'/g is analytic near b . Hence

$$\frac{f'(z)}{f(z)} - \frac{k}{z - b}$$

is analytic near b . Notice that k is positive if b was a zero, and negative if k was a pole, so

$$\frac{f'(z)}{f(z)} - \sum_j \frac{1}{z - z_j} + \sum_k \frac{1}{z - p_k}$$

is analytic in Ω_1 . Now integrate and use Cauchy's theorem.

Chapter 6 - Elementary Maps

Definition: (**Linear fractional transformation**)

An **LFT** is a function of the form

$$\frac{az - b}{cz - d}$$

for $ad - bc \neq 0$.

↳ Motivation: These are a simple but powerful class of functions. For example, the automorphisms $\mathbb{D} \rightarrow \mathbb{D}$ are types of LFTs. The automorphisms on S^2 are precisely the LFTs.

Theorem:

LFTs map circles onto circles and disks onto disks.

↳ Motivation: Useful when constructing conformal maps, since this makes it easy to transform disk-like regions.

Definition: (**Cayley transform**)

The **Cayley transform** is $\frac{z - i}{z + i}$.

↳ Motivation: It is a conformal map $\mathbb{H} \rightarrow \mathbb{D}$.

Theorem: (**LFTs can be prescribed three points**)

Given distinct points $z_1, z_2, z_3 \in \mathbb{C}^*$ and $w_1, w_2, w_3 \in \mathbb{C}^*$, there is a unique LFT T such that $T(z_j) = w_j$.

↳ Proof:

First assume the cross ratio works. Unicity: Fix $(0, 1, \infty)$.

If $w_1 = 0, w_2 = 1, w_3 = \infty$ then we can take

$$\frac{z - z_1}{z - z_3} \frac{z_2 - z_3}{z_2 - z_1}.$$

For the general case, we can take a map of the above type composed with its inverse, i.e.

$$T = \begin{matrix} & S^{-1} & \circ & R \\ (w_1, w_2, w_3) & \leftarrow & (0, 1, \infty) & \leftarrow & (z_1, z_2, z_3) \end{matrix}$$

Suppose there is another solution U . Then $V := S \circ U \circ R^{-1}$ maps $(0, 1, \infty) \mapsto (0, 1, \infty)$. Writing $V(z) = \frac{az+b}{cz+d}$ we see

$$\begin{aligned} 0 = V(0) &= \frac{b}{d} \implies b = 0, \\ \infty = V(\infty) &= \frac{a}{c} \implies c = 0, \\ 1 = V(1) &= \frac{a}{d} \implies V(z) = \frac{1z + 0}{0z + 1}, \end{aligned}$$

hence $V = \text{id}$ so $S^{-1} \circ R = U$.

Definition: (**Principal branch of log**)

The **principal branch** of the logarithm is given by

$$\log z = \log|z| + i \arg z$$

where $\arg z \in (-\pi, \pi]$.

Definition: (**Joukowski map**)

The **Joukowski map** is $w(z) := \frac{1}{2} \left(z + \frac{1}{z} \right)$.

↳ Motivation: Used in airfoil design. Also, circles at 0 are mapped to ellipses.

Chapter 7 - Harmonic Functions

Definition: (**Harmonic**)

A continuous function $u : \Omega \rightarrow \mathbb{R}$ is **harmonic** if it has the mean-value property. That is, for each $z \in \Omega$ there exists $r_z > 0$ such that $0 < r < r_z$ implies

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{it}) \, dt.$$

↳ Motivation: A priori, this generalizes those functions satisfying $\Delta u = 0$.

Definition: (**Subharmonic**)

A continuous function $u : \Omega \rightarrow [-\infty, +\infty)$ is **subharmonic** if it satisfies the mean-value inequality. That is, for each $z \in \Omega$ there exists $r_z > 0$ such that $0 < r < r_z$ implies

$$u(z) \leq \frac{1}{2\pi} \int_0^{2\pi} u(z + re^{it}) \, dt.$$

↳ Motivation: If f is analytic then $|f|$ and $\log|f|$ are subharmonic.

Theorem: (**Maximum principle**)

If u is subharmonic on Ω and attains its supremum, then u is constant.

↳ Proof:

Show $u^{-1}\{z_0\}$ is closed and open.

Suppose $u(z_0) = \sup u$. Let $E := u^{-1}\{u(z_0)\}$. Then E is nonempty and closed by continuity. If $z_1 \in E$, then

$$0 \leq \frac{1}{2\pi} \int_0^{2\pi} \underbrace{u(z_1 + re^{it}) - u(z_1)}_{\leq 0} \, dt$$

so $u(z_1 + re^{it}) = u(z_1)$ for sufficiently small r . Hence we found a disk in E , showing that E is open. We conclude that $E = \Omega$ by connectedness.

Theorem: (**Solution of Dirichlet problem on $\mathbb{D}$**)

Let $g : \partial\mathbb{D} \rightarrow \mathbb{R}$ be continuous. Then

$$u(z) := \text{PI}(g) = \frac{1}{2\pi} \int_0^{2\pi} \text{Re} \left(\frac{e^{it} + z}{e^{it} - z} \right) g(e^{it}) \, dt$$

solves the Dirichlet problem, i.e. u is harmonic on $\mathbb{D}$ and $\lim_{z \rightarrow \zeta \in \partial\mathbb{D}} u(z) = g(\zeta)$.

Definition: (**Poisson kernel**)

The **Poisson kernel** is

$$P_z(t) := \frac{1}{2\pi} \operatorname{Re} \left(\frac{e^{it} + z}{e^{it} - z} \right).$$

Definition: (**Herglotz kernel**)

The **Herglotz kernel** is

$$\frac{e^{it} + z}{e^{it} - z}.$$

↳ Motivation: Integrating against the Herglotz kernel with u harmonic gives an analytic f such that $\operatorname{Re} f = u$.

Theorem: (**Cauchy-Riemann equations**)

Suppose u, v are real-valued and continuously differentiable on Ω . Then, $u + iv$ is analytic if and only if

$$\begin{cases} u_x = v_y, \\ u_y = -v_x. \end{cases}$$

Definition: (**Harmonic conjugate**)

If $f = u + iv$, then v is a **harmonic conjugate** of u .

Theorem: (**Lindelöf's maximum principle**)

Suppose Ω is a region and $\{\zeta_1, \dots, \zeta_n\}$ is a subset of $\partial\Omega$ not equal to $\partial\Omega$. Suppose u is subharmonic and bounded on Ω . If

$$\limsup_{z \rightarrow \zeta} u(z) \leq m$$

for all $\zeta \in \Omega \setminus \{\zeta_1, \dots, \zeta_n\}$, then $u \leq m$ on Ω .

↳ Proof:

$$\text{Use } u(z) + \varepsilon \sum_{j=1}^n \log \left| \frac{z - \zeta_j}{d} \right|.$$

First suppose Ω is bounded and let $d := \operatorname{diam}(\Omega)$. For $\varepsilon > 0$ let

$$u_\varepsilon(z) := u(z) + \varepsilon \underbrace{\sum_{j=1}^n \log \left| \frac{z - \zeta_j}{d} \right|}_{\leq 0}.$$

We see that u_ε is subharmonic on Ω , and $u_\varepsilon \leq u$, and $\lim_{z \rightarrow \zeta_j} u_\varepsilon(z) = -\infty$. Hence $\limsup_{z \rightarrow \partial\Omega} u_\varepsilon(z) \leq m$. By the maximum principle, $u_\varepsilon \leq m$ on Ω . Letting $\varepsilon \rightarrow 0$ yields $u \leq m$.

Suppose instead that Ω is unbounded. Assume $\zeta_j \neq \infty$, possibly composing with an LFT. This means that $\limsup_{z \rightarrow \infty} u(z) \leq m$. Hence for $\varepsilon > 0$, there exists $R > \max |\zeta_j|$ and $u(z) \leq m + \varepsilon$ for $z \in \Omega \cap \{|z| \geq R\}$. Applying the bounded case to $u - \varepsilon$ on $\Omega \cap \{|z| < R\}$ gives $u - \varepsilon \leq m$ on $\Omega \cap \{|z| < R\}$. This is true for arbitrarily large R , so $u - \varepsilon \leq m$ on Ω . Finally let $\varepsilon \rightarrow 0$.

Theorem: (**Harnack's inequality**)

Let u be positive and harmonic on $\mathbb{D}$. Then for $|z| = r < 1$,

$$\frac{1-r}{1+r}u(0) \leq u(z) \leq \frac{1+r}{1-r}u(0).$$

↳ Motivation: Controls growth and decay on $\mathbb{D}$. Also used in the proof of Harnack's principle.

Theorem: (**Harnack's principle**)

Suppose $\{u_n\}$ is a monotonic increasing sequence of harmonic functions on Ω . Then either

(i) $\lim_{n \rightarrow \infty} u_n$ exists and is harmonic on Ω , or

(ii) $\lim_{n \rightarrow \infty} u_n \equiv +\infty$.

In both cases, the convergence is uniform on compact subsets.

Chapter 8 - Conformal Maps and Harmonic Functions

Theorem: (**Schwarz reflection principle**)

Suppose Ω is symmetric about $\mathbb{R}$. Denote $\Omega^\pm := \Omega \cap \{\operatorname{Im} z \gtrless 0\}$. Suppose v is harmonic on Ω^+ , continuous on $\Omega^+ \cup (\Omega \cap \mathbb{R})$. Then

$$V(z) := \begin{cases} v(z), & z \in \Omega \setminus \Omega^-, \\ -v(\bar{z}), & z \in \Omega^- \end{cases}$$

is harmonic on Ω . If also $v = \operatorname{Im} f$ where f is analytic on Ω^+ then

$$g(z) := \begin{cases} f(z), & z \in \Omega^+, \\ \overline{f(\bar{z})}, & z \in \Omega^- \end{cases}$$

extends to be analytic in Ω .

↳ Motivation: Useful to construct conformal maps between symmetric regions.

Definition: (**One-sided arc, two-sided arc**)

Let $\gamma : (0, 1) \rightarrow \mathbb{C}$ be an analytic arc, i.e. γ is injective and there is a conformal g on a neighborhood N of $(0, 1)$ such that $\gamma = g|_{(0,1)}$. Suppose next that the image of γ is contained in some $\partial\Omega$. We say that γ is **one-sided** if g, N can be taken such that $g(N \cap \mathbb{H}) \subseteq \Omega$ and $g(N \cap \overline{\mathbb{H}}) \subseteq \Omega \setminus \mathbb{C}$. We say that γ is **two-sided** if g, N can be taken such that $g(N \setminus (0, 1)) \subseteq \Omega$.

↳ Motivation: One-sided: Ω lies only on one side of $\gamma \subseteq \partial\Omega$. Two-sided: Ω lies on both sides of $\gamma \subseteq \partial\Omega$.

Theorem: (**Schwarz-Christoffel**)

Suppose $\partial\Omega$ is a polygon with vertices $v_1, \dots, v_n$ whose tangent direction increases by $\pi\alpha_j$ at v_j . Then there exists $x_1 < \dots < x_n$ and constants c_1, c_2 such that

$$f(z) = c_1 \int_{\gamma_z} \prod_{j=1}^n \frac{1}{(\zeta - x_j)^{\alpha_j}} d\zeta + c_2$$

is a conformal map $\mathbb{H} \rightarrow \Omega$, where γ_z is any curve from i to z .

↳ Motivation: Useful to construct conformal maps involving polygonal regions.

Chapter 9 - Calculus of Residues

Definition: (**Residue**)

Let f be analytic on an annulus at a . The **residue** of f at a , written $\text{Res}_{z=a} f(z)$, is the coefficient of $(z-a)^{-1}$ in the Laurent expansion of f at a .

Theorem: (**Residue theorem**)

Suppose f is analytic in Ω except for isolated singularities $a_1, \dots, a_n$, and that γ is a cycle with $\gamma \sim 0$ in Ω , and γ does not cross any singularities. Then

$$\frac{1}{2\pi i} \int_{\gamma} f(z) \, dz = \sum_{k=1}^n n(\gamma, a_k) \text{Res}_{a_k} f.$$

↳ Proof:

$\gamma \sim$ sum of circles; find primitive of most of the Laurent series.

Let D_k be a small disk at a_k . Then $\gamma \sim \sum_k n(\gamma, a_k) \cdot \partial D_k$ in $\Omega \setminus \{a_k\}$. Therefore,

$$\frac{1}{2\pi i} \int_{\gamma} f(z) \, dz = \sum_{k=1}^n n(\gamma, a_k) \frac{1}{2\pi i} \int_{\partial D_k} f(z) \, dz.$$

Write the Laurent series of f at a_k as $f(z) = \sum_n b_n(z-a_k)^n$. Then

$$\begin{aligned} \frac{1}{2\pi i} \int_{\partial D_k} f(z) \, dz &= \frac{1}{2\pi i} \int_{\partial D_k} \frac{b_{-1}}{z-a_k} \, dz + \frac{1}{2\pi i} \int_{\partial D_k} \left(\dots + \frac{b_{-2}}{(z-a_k)^2} + b_0(z-a_k)^0 + \dots \right) \, dz \\ &= b_{-1} + \frac{1}{2\pi i} \int_{\partial D_k} \frac{d}{dz} \left(\dots + \frac{-b_{-2}}{(z-a_k)^1} + b_0(z-a_k)^1 + \dots \right) \, dz = b_{-1} = \text{Res}_{a_k} f. \end{aligned}$$

Inserting this in the above completes the proof.

Chapter 10 - Normal Families

Definition: (**Normal family**)

A family $\mathcal{F}$ of continuous functions on Ω is **normal** if every sequence $\{f_n\}$ has a subsequence which converges uniformly on compact subsets.

Definition: (**Equicontinuous, uniformly equicontinuous**)

A family $\mathcal{F}$ on a set $E \subseteq \mathbb{C}$ is

- (a) **equicontinuous** at $z_0 \in E$ if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $|z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon$ for all $f \in \mathcal{F}$;
- (b) **equicontinuous** on E if it is equicontinuous at all points in E ;
- (c) **uniformly equicontinuous** if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $|z - z_0| < \delta \implies |f(z) - f(z_0)| < \varepsilon$ for all $z_0 \in E$ and $f \in \mathcal{F}$.

Theorem: (**Arzela-Ascoli**)

A family $\mathcal{F} \subseteq C(\Omega)$ is normal iff

- (i) $\mathcal{F}$ is equicontinuous on Ω , and
- (ii) there exists $z_0 \in \Omega$ such that $\{f(z_0) : f \in \mathcal{F}\}$ is bounded.

Definition: (**Locally bounded**)

A family $\mathcal{F} \subseteq C(\Omega)$ is **locally bounded** if each $w \in \Omega$ lies in a small disk on which $\mathcal{F}$ is bounded.

Theorem: (**Arzela-Ascoli for analytic functions**)

For a family $\mathcal{F}$ of analytic functions on Ω , the following are equivalent:

- (i) $\mathcal{F}$ is normal on Ω
- (ii) $\mathcal{F}$ is locally bounded on Ω
- (iii) $\mathcal{F}'$ is locally bounded on Ω , and there exists $z_0 \in \Omega$ such that $\{f(z_0) : f \in \mathcal{F}\}$ is bounded.

↳ Proof:

(i) $\implies$ (ii):

Use Arzela-Ascoli.

By Arzela-Ascoli, $\mathcal{F}$ is equicontinuous and there exists $z_0 \in \Omega$ such that $\mathcal{F}(z_0)$ is bounded. By the proof of Arzela-Ascoli, we are done.

(ii) $\implies$ (iii):

Use Cauchy's estimate.

For $z_0 \in \Omega$, there exists a closed disk $\overline{D}$ of radius r at z_0 such that $|\mathcal{F}(\overline{D})| \leq M$. By Cauchy's estimate, $|f'(\overline{D})| \leq \frac{1}{r}M$.

(iii) $\implies$ (i):

Integrate to show that $\mathcal{F}$ is equicontinuous.

Fix $z_0 \in \Omega$. For z near z_0 and $f \in \mathcal{F}$, we have $|f'(z)| \leq M$. By integrating,

$$|f(z) - f(z_0)| = \left| \int_{z_0}^z f' \right| \leq \int_{z_0}^z |f'| \leq M|z - z_0|.$$

This shows that $\mathcal{F}$ is equicontinuous, so by Arzela-Ascoli we are done.

Theorem: (**Riemann mapping theorem**)

Suppose $\Omega \subset \mathbb{C}$ is simply connected and $\Omega \neq \mathbb{C}$ and pick $z_0 \in \Omega$. Then there exists a unique bijective analytic map $f : \Omega \rightarrow \mathbb{D}$ such that $f(z_0) = 0$ and $f'(z_0) > 0$.

↳ Proof:

Use local boundedness $\implies$ normality of $\mathcal{F} = \{f : \Omega \rightarrow \mathbb{D} : \text{analytic, injective, } f(z_0) = 0, f'(z_0) > 0\}$.

Uniqueness was shown earlier. Let $\mathcal{F} := \{f : \Omega \rightarrow \mathbb{D} : \text{analytic, injective, } f(z_0) = 0, f'(z_0) > 0\}$. We showed earlier that $\mathcal{F}$ is nonempty. Notice that $\mathcal{F}$ is bounded and in particular locally bounded, so Arzela-Ascoli gives that $\mathcal{F}$ is normal. Take a sequence $\{f_n\}$ such that $\lim_{n \rightarrow \infty} f_n'(z_0) = \sup_{f \in \mathcal{F}} f'(z_0) =: M$. By normality we may replace $\{f_n\}$ with a subsequence which converges uniformly on compact subsets. Denote the limit as f .

Claim: $f \in \mathcal{F}$. By Weierstrass' theorem, f is analytic and $f(z_0) = 0$ and $f'(z_0) = M$. Notice that $M > 0$ since $\mathcal{F}$ has injective functions, so f is nonconstant and hence injective by Hurwitz's theorem.

Claim: f surjects onto $\mathbb{D}$.

Move ζ_0 to 0, then take $\sqrt{\cdot}$, then move the image of z_0 to 0. This composed with its "inverse" recovers f .

Suppose there exists $\zeta_0 \in \mathbb{D} \setminus f(\Omega)$. Let $g_1 := T_{0 \leftarrow \zeta_0} \circ f$. Then g_1 cannot attain 0, and since $g_1(\Omega)$ is simply connected we can define $g_2 := \sqrt{g_1}$ to be injective and analytic. Let $g := T_{0 \leftarrow g_2(z_0)} \circ g_2$. Writing $\lambda := \frac{|g'(z_0)|}{g'(z_0)}$, we have $\lambda g \in \mathcal{F}$. Let

$$\varphi := T_{0 \leftarrow \zeta_0}^{-1} \circ z^2 \circ T_{0 \leftarrow g_2(z_0)}^{-1}.$$

We notice $f = \varphi \circ g$. By Schwarz's lemma, $|\varphi'(0)| < 1$ (noting φ is 2-1 and hence not a rotation), so

$$|f'(z_0)| = |\varphi'(0)| \cdot |g'(z_0)| < |g'(z_0)| = \lambda g'(z_0)$$

contradicting maximality.

Chapter 11 - Series and Products

Definition: (**Singular part**)

Let f be meromorphic near z_0 . The **singular part** of f at b , written $S_b(f)$, is the part of the Laurent series of f at z_0 containing negative powers.

Theorem: (**Mittag-Leffler**)

Suppose $\{b_k\}$ is a sequence of distinct points such that $b_k \rightarrow \partial\Omega$ as $k \rightarrow \infty$. Set

$$S_k(z) := \sum_{j=1}^{n_k} \frac{c_{j,k}}{(z - b_k)^j}.$$

Then there exists a meromorphic function f on Ω with singular part S_k at b_k .

↳ Proof:

Approximate $\sum_{b_k \in K_{n+1} \setminus K_n} S_k$ by Runge's theorem, then sum.

Let $K_n := \{z \in \Omega : \text{dist}(z, \partial\Omega) \geq \frac{1}{n} \text{ and } |z| \leq n\}$ be monotone increasing compact sets growing to Ω . Each K_n contains finitely many points b_k . By Runge's theorem, there exists a rational function r_n with poles outside Ω such that

$$\left| \sum_{b_k \in K_{n+1} \setminus K_n} S_k(z) - r_n(z) \right| < 2^{-n} \quad \forall z \in K_n.$$

Therefore for each m ,

$$\sum_{n \geq m} \left(\sum_{b_k \in K_{n+1} \setminus K_n} S_k(z) - r_n(z) \right)$$

converges uniformly on K_m by Weierstrass' M -test, and is hence analytic on K_m by Weierstrass' theorem. Set

$$f(z) := \sum_{b_k \in K_1} S_k(z) + \sum_{n=1}^{\infty} \left(\sum_{b_k \in K_{n+1} \setminus K_n} S_k(z) - r_n(z) \right)$$

which converges uniformly on compact subsets of $\Omega \setminus \{b_1, \dots\}$ by the above argument, and is hence analytic on $\Omega \setminus \{b_1, \dots\}$ by Weierstrass' theorem. The singular parts are as desired.

Theorem: (**Weierstrass product theorem**)

Suppose Ω is bounded, and $\{b_j\}$ is a sequence in Ω such that $b_j \rightarrow \partial\Omega$, and $\{n_j\}$ are positive integers. Then there exists an analytic function on Ω with zeros of order n_j at precisely b_j .

Theorem: (**Jensen's formula**)

Suppose f is meromorphic on $|z| \leq R$ with zeros $\{a_k\}$ and poles $\{b_k\}$, and suppose that 0 is not a zero or pole of f . Then

$$\frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{it})| = \log |f(0)| + \sum_{|a_k| < R} \log \frac{R}{|a_k|} - \sum_{|b_j| < R} \log \frac{R}{|b_j|}.$$

↳ Motivation: Helps control the growth of the zero set.

Chapter 13 - The Dirichlet Problem

Definition: (**The Dirichlet Problem**)

The **Dirichlet problem** for a function $f \in C(\partial\Omega)$ is to find a harmonic function u on Ω which extends continuously to agree with f on $\partial\Omega$.

Definition: (**Perron family**)

A family $\mathcal{F}$ of subharmonic functions on Ω is a **Perron family** if

- (i) $v_1, v_2 \in \mathcal{F} \implies \max(v_1, v_2) \in \mathcal{F}$;
- (ii) for all $v \in \mathcal{F}$ and disks D with $\overline{D} \subseteq \Omega$ and $v > -\infty$ on ∂D , we may replace v with $\text{PI}(v)$ inside D and get another function in $\mathcal{F}$;
- (iii) for all $z \in \Omega$ there exists $v \in \mathcal{F}$ such that $v(z) > -\infty$.

↳ Motivation: To solve Dirichlet problems, we take $\mathcal{F}_f := \{v : \text{subharmonic, } \limsup_{z \in \Omega} v(z) \leq f(\zeta) \ \forall \zeta \in \partial\Omega\}$.

Definition: (**Perron solution**)

The **Perron solution** of the Dirichlet problem on Ω for f is

$$u_f := \sup_{v \in \mathcal{F}_f} v.$$

↳ Motivation: It is always harmonic on Ω ,

Definition: (**Local barrier**)

Consider $\zeta_0 \in \partial\Omega$. We say that b is a **local barrier** at ζ_0 if for some disk D at ζ_0 , we have that $b : \Omega \cap D \rightarrow [-\infty, 0)$ is subharmonic and $\lim_{z \rightarrow \zeta_0} b(z) = 0$.

Definition: (**Regular point, etc.**)

A point $\zeta_0 \in \partial\Omega$ is a **regular point** if it has a local barrier, otherwise it is an **irregular point**. If all points on $\partial\Omega$ are regular, then Ω is a **regular region**.

↳ Motivation: The Perron solution has the correct boundary value at regular points ζ_0 , given continuity and boundedness of the boundary function f .